Fiscal Year:	FY 2025	**Task Last Updated:**	FY 06/27/2025
PI Name:	Marin de Evsikova, Caralina Ph.D.		
Project Title:	Stressors to Spaceflight: Identification of Transposon-Driven Changes to Gene Networks in GeneLab Data		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate (3) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	33744-8200	**Congressional District:**	13
Comments:	PI name is listed as de Evsikova in some publications. It is listed as Evsikova on the NSSC website.		
Project Type:	Ground	**Solicitation / Funding Source:**	2021 Space Biology NNH21ZDA001N-SBAS E.11: Animal Studies
Start Date:	09/01/2024	**End Date:**	08/31/2026
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:	Jones, Harry	**Contact Phone:**	650.604.5518
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 08/31/2026 per F. Hernandez/NASA-ARC (Ed., 9/17/25). NOTE: Period of performance (POP) corrected per F. Hernandez/NASA-ARC. POP is 09/01/2024 - 08/31/2025. Previous POP was listed as 7/24/24 - 7/24/25. (Ed., 2/12/25).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Evsikov, Alexei (Bay Pines Foundation)		
Grant/Contract No.:	80NSSC24K1807		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Here, we propose to test the novel hypothesis that transposons act as a molecular driver for remodeling gene networks caused by stress and adaptation responses to the extreme environments of spaceflight across species, ranging from simpler invertebrates to more complex mammals. Using our innovative Transcript and Transposons Expression Signature Analysis (TTESA) pipeline – to detect and quantify altered gene networks' transposable element (TE) expression after space exposure – will provide data critical to understand the impact of TEs on neighboring gene expression as a potential mechanism remodeling genetic networks at both the genomic and transcriptomic levels. Albeit TEs are not considered to be reactivated from exposure to spaceflight, our preliminary analysis with medaka fish exposed to low Earth orbit (LEO) have up to 250X increase in some TEs, and we have the first evidence spaceflight induces TE-driven gene transcription in mice. It is unknown if spaceflight-induced TE-driven genes are permanent upon return to Earth. TEs are an unrecognized and unmitigated potential risk factor for healthy space exploration, given their established roles in reproduction, stem cells, and cancer.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work investigates the potential health impacts of spaceflight on genome stability, altered gene networks, and gene expression as potential risk factors for subsequent disease in veteran astronauts. The International Space Station (ISS) has maintained a continuous human presence in orbit for more than 20 years, and microgravity research on space stations has provided valuable insight for future long-term human missions in space. As of late 2024 690 astronauts have journeyed to space, with the majority from military service. Veteran astronauts typically complete multiple missions in space over their careers. The impacts of sustained and multiple spaceflights on the genomes and subsequent health and disease in these veterans are unknown. Retired astronauts represent a unique group of veterans, which is anticipated grow exponentially with the next generation from the recently formed US Space Force and advent of commercial spaceflight. By investigating these health outcomes in the unique context of spaceflight exposure, this research has the potential to uncover novel biological mechanisms and risk factors. Altered gene expression via transponson activation is unrecognized and unmitigated potential risk factor for healthy space exploration, given their established roles in reproduction, stem cells, and cancer.
Task Progress:	Using our bioinformatics pipeline to test our hypothesis that stress from adaptation to gravitation encountered on the Moon or Mars alters expression of TEs and gene networks for developmental systems in larval and adult flies (D. melanogaster). Results: Flies exposed to either microgravity or simulated hypergravity altered multiple developmental systems; most robustly, metabolism, by altering gene networks underlying mitochondrial biogenesis, anatomy, replication, oxidative phosphorylation, and fat metabolism. Furthermore, multiple gene networks underlying growth and reproduction were universally suppressed, whereas genes regulating stress and immune responses were robustly activated after microgravity and hypergravity. Some TEs were substantially altered during adaptation to microgravity (8-100X). TEs have strong effects on fruit flies, particularly fertility and metabolism. These experiments facilitate understanding the role of TEs in development by inducing robust transcription of TEs normally not observed on Earth, and serve as translational models for humans living under altered gravitational environments, such as astronauts and retired veteran astronauts.
Bibliography Type:	Description: (Last Updated: 06/27/2025)
Abstracts for Journals and Proceedings	Evsikov AV, Evsikov RA, Evsikova EA, Marin de Evsikova C. "From Moon to Martian gravity: Detecting developmental mechanisms in larval and adult flies adapting to extreme environments in space." 84th Annual Meeting for the Society for Developmental Biology, San Juan, Puerto Rico, June 18-22, 2025. Abstracts. 84th Annual Meeting for the Society for Developmental Biology, San Juan, Puerto Rico, June 18-22, 2025. , Jun-2025
Abstracts for Journals and Proceedings	Evsikov AV, Marin de Evsikova C. "Bioinformatics to detect transposon-driven gene expression changes from spaceflight in C. elegans. " 9th Annual Florida Worm Meeting, Melbourne, FL, May 24, 2025. Abstracts. 9th Annual Florida Worm Meeting, Melbourne, FL, May 24, 2025. , May-2025
Abstracts for Journals and Proceedings	Marin de Evsikova C. "Bioinformatics to detect transposon and gene expression changes from spaceflight and low Earth orbit: Implications for veteran astronaut health." 2025 National VA Research Week 2025, Bay Pines, FL, May 12-16, 2025. Abstracts. 2025 National VA Research Week 2025, Bay Pines, FL, May 12-16, 2025. , May-2025